

V BOOPATHY KHANISHK

Set 7 : Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID Age Gender Income (in \$)

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

SOLUTION :

CODE :

01.

%%R

```
library(ggplot2)
```

```
df7 <- data.frame(  
  Customer_ID = factor(c(1, 2, 3)),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)
```

```
ggplot(df7, aes(x = Customer_ID, y = Age, fill = Customer_ID)) +  
  geom_bar(stat = "identity", width = 0.5) +  
  geom_text(aes(label = paste0(Age, " yrs")), vjust = -0.5, size = 4, fontface =  
"bold") +  
  labs(  
    title = "Customer Age Distribution",  
    x = "Customer ID",  
    y = "Age (Years)",  
    fill = "Customer ID"  
  ) +  
  theme_minimal() +  
  scale_fill_brewer(palette = "Blues")
```

02. %%R

```
library(ggplot2)
```

```
library(dplyr)
```

```
df7 <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)
```

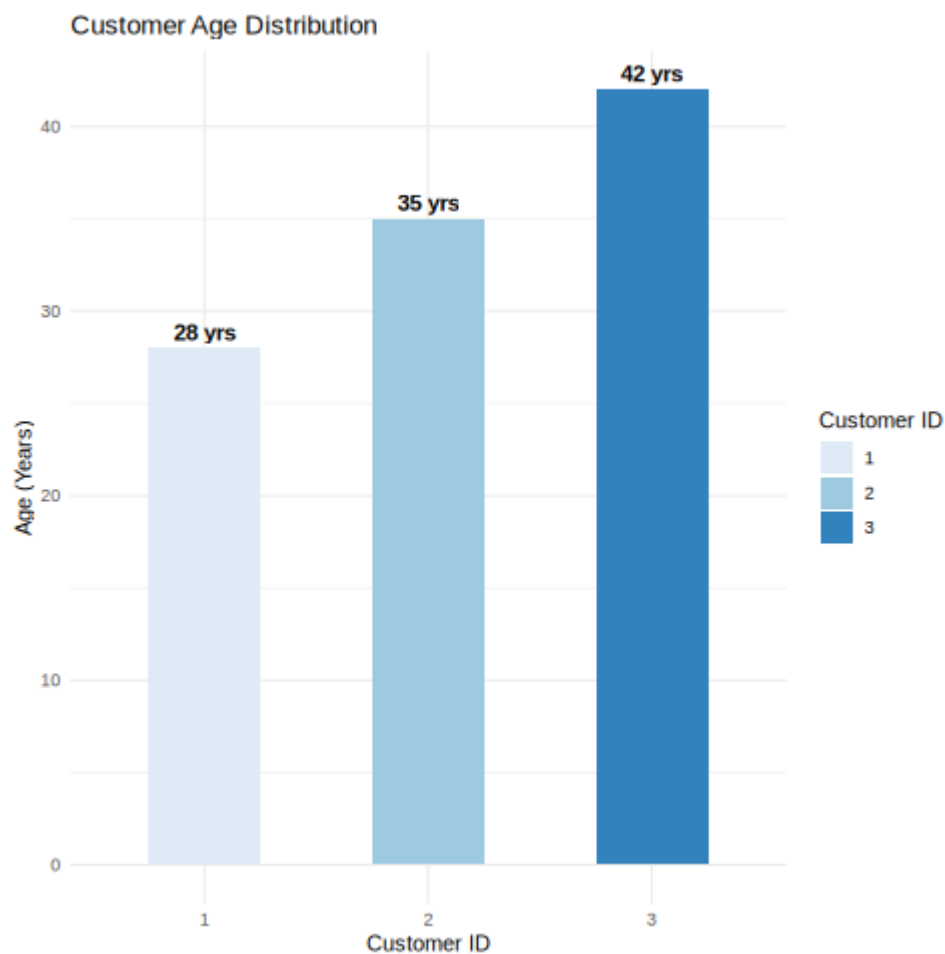
```
gender_df <- df7 %>%  
  group_by(Gender) %>%  
  summarise(Count = n()) %>%  
  mutate(Percentage = round((Count / sum(Count)) * 100, 1))
```

```
ggplot(gender_df, aes(x = "", y = Count, fill = Gender)) +  
  geom_bar(stat = "identity", width = 1) +  
  coord_polar("y", start = 0) +  
  geom_text(  
    aes(label = paste0(Gender, "\n", Count, " (", Percentage, "%)")),  
    position = position_stack(vjust = 0.5),  
    color = "white",  
    fontface = "bold",  
    size = 4.5  
  ) +  
  labs(  
    title = "Customer Gender Distribution",  
    x = NULL,  
    y = NULL,  
    fill = "Gender"  
  ) +  
  theme_void() +  
  scale_fill_manual(values = c("Female" = "#E41A1C", "Male" = "#377EB8"))
```

03. %%R

```
df7 <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income_USD = c(50000, 60000, 75000)  
)  
  
df7_table <- df7  
df7_table$Income_USD <- paste0("$", format(df7_table$Income_USD,  
big.mark = ","))  
  
print("--- CUSTOMER DEMOGRAPHICS TABLE ---")  
print(df7_table)
```

OUTPUT :



Customer Gender Distribution

